A

Major Project

On

A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads

(Submitted in partial fulfilment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

P Bhavani Priya (227R1A05Q1)

Valaboju Nithin (227R1A05R1)

G Sriram (227R1A05M9)

Under the Guidance of

D Sandhya Rani

(Assistant Professor)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "**A Data – Intelligent Framework For Detecting Manipulated Click In Online Ads**" being submitted by **P Bhavani Priya(227R1A05Q1), Valaboju Nithin (227R1A05R1) & G Sriram (227R1A05M9)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

D Sandhya Rani
Assistant Professor
INTERNAL GUIDE

Dr. Nuthanakanti Bhaskar
HoD

Dr. A. Raji Reddy
DIRECTOR

Signature of External Examiner

Submitted for viva voice Examination held on _____

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P Bhavani Priya (227R1A05Q1)
Valaboju Nithin (227R1A05R1)
G Sriram (227R1A05M9)

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ABSTRACT

Online advertising has become a major source of revenue for digital platforms. However, it is increasingly threatened by ad click fraud, where malicious users or automated bots generate fake clicks to manipulate advertising metrics or increase costs for advertisers. Traditional rule-based detection systems are often ineffective in handling evolving and sophisticated fraud techniques. This project presents an intelligent ad click fraud detection system using Machine Learning and Deep Learning algorithms to accurately distinguish between genuine and fraudulent clicks. The system analyses user behaviour patterns, traffic characteristics, and temporal features such as click frequency, session duration, and time intervals between clicks.

Various machine learning models, including Random Forest and Support Vector Machine, along with deep learning models such as Neural Networks, are implemented to improve detection performance. The proposed system enhances accuracy, reduces false positives, and adapts to new fraud patterns over time. Overall, this approach helps in minimizing financial losses, improving the reliability of online advertising systems, and increasing trust among advertisers and digital platforms.

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Keywords- Ad Click Fraud Detection, Machine Learning, Deep Learning, Clickstream Analysis, Behavioural Analysis, Feature Engineering, Classification Models, Real-Time Fraud Detection, Digital Advertising Security, Bot Detection.

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
Figure 3.1	System Architecture Of A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads	15
Figure 3.2	Dataflow Diagram of A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads	18
Figure 5.1	GUI/Login Interface of the system	34
Figure 5.2	Admin dashboard showing remote users information	35
Figure 5.3	Dataset upload interface of the system .	35
Figure 5.4	Dataset details view interface of the system	36

Figure 5.5	Model training results interface of the system	37
Figure 5.6	Model accuracy bar chart interface of the system	37
Figure 5.7	Model accuracy line chart interface of the system	38
Figure 5.8	Prediction ratio interface of ad click fraud detection system	38
Figure 5.9	User registration interface of the system	39
Figure 5.10	User login interface of the system	40
Figure 5.11	Input interface for ad click fraud detection system	40
Figure 5.12	Prediction result interface of ad click fraud detection system	41

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
Table 6.2.1	Uploading Dataset	44
Table 6.2.2	Classification (Fraud detection)	44

TABLE OF CONTENTS

ABSTRACT				i
LIST OF FIGURES				ii
LIST OF TABLES				v
1.	**INTRODUCTION**			2
	1.1	PROJECT PURPOSE		2
	1.2	PROJECT FEATURES		3
2.	**LITERATURE SURVEY**			6
	2.1	REVIEW OF RELATED WORK		7
	2.2	DEFINITION OF PROBLEM STATEMENT		9
	2.3	EXISTING SYSTEM		9
	2.4	PROPOSED SYSTEM		11
	2.5	OBJECTIVES		12
	2.6	HARDWARE & SOFTWARE REQUIREMENTS		13
		2.6.1	HARDWARE REQUIREMENTS	13
		2.6.2	SOFTWARE REQUIREMENTS	13
3.	**SYSTEM ARCHITECTURE & DESIGN**			15
	3.1	PROJECT ARCHITECTURE		15
	3.2	DESCRIPTION		16
	3.3	DATA FLOW DIAGRAM		17
4.	**IMPLEMENTATION**			20
	4.1	ALGORITHMS USED		20
	4.2	SAMPLE CODE		24
5.	**RESULTS & DISCUSSION**			34
6.	**VALIDATION**			43
	6.1	INTRODUCTION		43
	6.2	TEST CASES		44
		6.2.1	UPLOADING DATASET	44
		6.2.2	CLASSIFICATION	44
7.	**CONCLUSION & FUTURE ASPECTS**			46
	7.1	PROJECT CONCLUSION		46
	7.2	FUTURE ASPECTS		47
8.	**BIBLIOGRAPHY**			48
	8.1	REFERENCES		48
	8.2	GITHUB LINK		50
	8.3	PUBLICATIONS		50

1.INTRODUCTION

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1. INTRODUCTION

Online advertising has experienced rapid growth over the past decade and has become one of the primary revenue sources for digital platforms. Among various advertising models, the Pay-Per-Click (PPC) model is widely used, where advertisers are charged each time a user clicks on their advertisement. This model provides measurable performance and allows businesses to reach targeted audiences effectively. However, the increasing dependence on online advertising has also led to the rise of fraudulent activities, particularly ad click fraud. Ad click fraud refers to the generation of illegitimate clicks on online advertisements by malicious users, competitors, or automated bots. The main objective of such fraud is to increase advertising costs for competitors or to generate illegal revenue for publishers. These fraudulent clicks distort advertising analytics, reduce the return on investment (ROI) for advertisers, and negatively impact the credibility of digital marketing platforms.

Traditional fraud detection systems are mostly rule-based and rely on predefined conditions such as repeated clicks from the same IP address, unusual click frequency, or suspicious geographic locations. Although these methods are simple to implement, they are not capable of adapting to evolving fraud techniques. Modern fraudsters use sophisticated bots that mimic human behavior, making it difficult for static rule-based systems to detect them effectively. This results in high false positives and missed fraud detection.

To overcome these limitations, intelligent techniques based on Machine Learning (ML) and Deep Learning (DL) have emerged as powerful solutions. These approaches can analyse large volumes of data, learn patterns from historical click behavior, and automatically adapt to new fraud strategies. Machine learning algorithms such as Logistic Regression, Support Vector Machines, and Random Forest can identify patterns in structured data, while deep learning models like Neural Networks and Long Short-Term Memory (LSTM) networks can capture complex and time-dependent relationships in clickstream data.

1.1 PROJECT PURPOSE

The primary purpose of this project is to design and develop an intelligent system for detecting ad click fraud in online advertising platforms using Machine Learning and Deep Learning techniques. With the rapid growth of digital marketing and Pay-Per-Click (PPC) advertising models, the need for an efficient and reliable fraud detection system has become critical. Ad click fraud poses a serious threat to advertisers and publishers by generating illegitimate clicks through bots, scripts, or malicious users. These fraudulent activities lead to financial losses, inaccurate campaign analytics, and reduced trust in online advertising systems. Traditional rule-based detection methods are no longer sufficient to handle advanced and evolving fraud patterns.

This project aims to overcome these challenges by implementing a data-driven approach that can automatically learn and identify fraudulent patterns from large-scale ad click data. By analyzing user behavior, click frequency, session duration, and time-based patterns, the system is capable of distinguishing between genuine and fraudulent clicks with high accuracy.

Another important purpose of this project is to build a scalable and adaptable system that can continuously improve its performance over time. The integration of machine learning models such as Random Forest and deep learning models like Neural Networks ensures that the system can handle complex and non-linear relationships in the data.

1.2 PROJECT FEATURES

The proposed Ad Click Fraud Detection System incorporates several advanced features that enhance its efficiency, accuracy, and usability in detecting fraudulent advertisement clicks. These features are designed to overcome the limitations of traditional systems and provide a scalable and intelligent solution.

1. Intelligent Fraud Detection

The system uses Machine Learning and Deep Learning algorithms to automatically identify fraudulent click patterns. Unlike traditional rule-based systems, it learns from data and adapts to new and evolving fraud techniques.

2. Data Preprocessing Capability

The system includes a robust data preprocessing module that handles missing values, removes duplicate records, and converts categorical data into numerical format using encoding techniques. This ensures high-quality input data for model training.

3. Feature Extraction and Selection

Important behavioural features such as click frequency, session duration, time intervals between clicks, device type, and geographical patterns are extracted. These features help in accurately distinguishing between genuine and fraudulent clicks.

4. Multiple Model Implementation

The system supports both traditional Machine Learning models (such as Logistic Regression, Random Forest, and Support Vector Machine) and Deep Learning models (such as Neural Networks and LSTM). This improves overall detection performance.

5. Real-Time Fraud Detection

The system is capable of analyzing incoming click data and providing instant predictions. This enables real-time identification and prevention of fraudulent activities.

6. High Accuracy and Reduced False Positives

By using advanced algorithms and feature engineering, the system achieves high detection accuracy while minimizing false positives, ensuring that genuine users are not incorrectly flagged.

7. Scalable and Efficient System

The system is designed to handle large volumes of ad click data efficiently, making it suitable for real-world advertising platforms with high traffic.

8. User-Friendly Interface

A simple and interactive web interface is provided for users to upload datasets, train models, and view results. This makes the system easy to use even for non-technical users.

9. Model Evaluation and Reporting

The system evaluates model performance using metrics such as accuracy and displays results in a clear format. This helps in comparing different models and selecting the best one.

10. Secure and Reliable Operation

The system ensures secure user authentication and protects sensitive data. It is designed to operate reliably even under heavy workloads.

2.LITERATURE SURVEY

2. LITERATURE SURVEY

Ad click fraud detection has been an active area of research due to its significant impact on online advertising systems. Various researchers have proposed different machine learning and deep learning approaches to identify fraudulent clicks effectively. The following are some of the important studies related to this domain:

1. Click Fraud Detection in Online Advertising: A Machine Learning Approach

Nagaraja, A., Prasanna, R., and Kumar, S. proposed a machine learning-based approach to detect fraudulent clicks in online advertising systems. The study utilizes real-world clickstream datasets containing features such as IP address, click frequency, session duration, and device type. Various supervised learning algorithms, including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines, were implemented and compared.

The results indicated that ensemble methods like Random Forest achieved better performance in terms of accuracy and reduced false positives. The study also highlighted the importance of feature engineering and data preprocessing in improving fraud detection efficiency. This research demonstrates that machine learning techniques are scalable and effective for real-time fraud detection.

2. Detection of Click Fraud in Online Advertising Using Deep Neural Networks

Zhang, Y., Li, H., and Wang, X. introduced a deep learning-based framework for detecting ad click fraud using Deep Neural Networks (DNN). The model was trained on large-scale clickstream data with features such as timestamps, user navigation patterns, browser information, and geographic location.

The experimental results showed that deep learning models outperform traditional machine learning algorithms like SVM and Naïve Bayes, especially in detecting complex and low-frequency fraud patterns. The study concluded that deep neural networks are highly effective in learning non-linear relationships and adapting to evolving fraud strategies.

3. A Hybrid Machine Learning Model for Ad Click Fraud Detection

Patel, R., Mehta, D., and Shah, P. proposed a hybrid model that combines both supervised and unsupervised learning techniques. Initially, clustering algorithms such as K-Means were used

to group similar user behavior patterns and identify suspicious clusters. Then, classification models like Gradient Boosting and Random Forest were applied to classify clicks.

The hybrid approach improved detection accuracy and significantly reduced false alarms compared to standalone models. The study emphasized that combining multiple learning techniques enhances robustness and adaptability against evolving fraud patterns.

4. Click Fraud Detection Using Recurrent Neural Networks

Chen, L., Zhou, M., and Sun, Q. focused on detecting time-dependent fraud patterns using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models. Since fraudulent clicks often follow temporal patterns, the use of sequence-based models helps in identifying abnormal user behavior over time.

The results showed that LSTM-based models outperform traditional classifiers in detecting coordinated bot attacks and repeated click patterns. This research highlights the importance of temporal analysis in fraud detection systems.

5. An Intelligent Framework for Ad Click Fraud Detection Using Ensemble and Deep Learning Models

Singh, P., Verma, A., and Rao, K. proposed an intelligent framework that integrates ensemble learning techniques with deep learning models. The system combines algorithms such as Gradient Boosting, Random Forest, and Deep Neural Networks to improve overall detection performance.

The study demonstrated that combining multiple models leads to higher accuracy, better scalability, and improved robustness. It also showed that ensemble-deep learning approaches are highly effective in handling large-scale datasets and dynamic fraud scenarios.

2.1 REVIEW OF RELATED WORK

Ad click fraud detection has gained significant attention in recent years due to the rapid growth of online advertising platforms. Researchers have explored various techniques, including rule-based methods, machine learning algorithms, and deep learning approaches, to effectively identify fraudulent click activities.

Earlier systems primarily relied on rule-based detection mechanisms such as IP blocking, click thresholds, and blacklist filtering. These methods were simple and easy to implement but lacked adaptability. As fraudsters began using advanced techniques to mimic human behavior, these traditional approaches became less effective, resulting in higher false positives and undetected fraud cases.

To overcome these limitations, researchers introduced machine learning-based approaches for click fraud detection. Supervised learning algorithms such as Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM) have been widely used. These models analyze historical click data and learn patterns that differentiate legitimate clicks from fraudulent ones. Among these, ensemble methods like Random Forest have shown better performance due to their ability to handle large datasets and reduce overfitting.

In addition to traditional machine learning techniques, deep learning methods have also been applied to improve fraud detection accuracy. Deep Neural Networks (DNN) are capable of learning complex and non-linear relationships within data, making them suitable for detecting sophisticated fraud patterns. Furthermore, Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models have been used to analyze sequential and time-based click behavior. These models are particularly effective in identifying repeated and coordinated bot-driven attacks. Recent research has also focused on hybrid and ensemble approaches that combine multiple algorithms to enhance detection performance. By integrating clustering techniques with classification models or combining machine learning and deep learning methods, these approaches achieve higher accuracy and robustness. They also help in reducing false alarms and improving the overall reliability of the system.

Despite these advancements, challenges such as data imbalance, evolving fraud techniques, and real-time detection still exist. Therefore, there is a need for intelligent, scalable, and adaptive systems that can continuously learn from new data and improve their performance.

The proposed system builds upon these existing approaches by integrating both machine learning and deep learning techniques to provide a more accurate and efficient solution for ad click fraud detection.

2.2 DEFINITION OF PROBLEM STATEMENT

Online advertising, especially the Pay-Per-Click (PPC) model, has become a major source of revenue for digital platforms. However, it is highly vulnerable to ad click fraud, where malicious users, competitors, or automated bots generate fake clicks on advertisements. These fraudulent activities result in significant financial losses for advertisers, distort campaign performance metrics, and reduce trust in online advertising systems. Detecting ad click fraud is a challenging task due to several factors. Firstly, the volume of clickstream data generated in real-time is extremely large, making it difficult to process and analyze efficiently. Secondly, fraudsters continuously evolve their techniques by mimicking legitimate user behavior, making fraudulent clicks appear similar to genuine ones. This similarity increases the complexity of accurately distinguishing between valid and invalid clicks.

Traditional rule-based detection systems, which rely on predefined conditions such as repeated clicks from the same IP address or abnormal click frequency, are no longer sufficient. These systems lack adaptability and fail to detect sophisticated and large-scale fraud attacks. They often produce high false positives, where legitimate user clicks are incorrectly flagged as fraudulent, thereby affecting user experience and advertising performance. Another major challenge is the presence of imbalanced datasets, where the number of legitimate clicks significantly outweighs fraudulent clicks. This imbalance makes it difficult for detection models to learn and accurately identify fraud patterns. Additionally, the need for real-time detection further increases the complexity, as the system must provide fast and accurate predictions without delays.

Therefore, the problem addressed in this project is to develop an intelligent, scalable, and adaptive ad click fraud detection system that can efficiently analyze large volumes of data, accurately identify fraudulent clicks, minimize false positives, and adapt to evolving fraud patterns using advanced Machine Learning and Deep Learning techniques.

2.3 EXISTING SYSTEM

The existing ad click fraud detection systems primarily rely on traditional rule-based techniques to identify fraudulent activities. These systems use predefined rules and conditions such as repeated clicks from the same IP address, unusually high click frequency, abnormal

geographic locations, and blacklisted users to detect suspicious behaviour. In many cases, these systems implement threshold-based monitoring, where a certain number of clicks within a short period is considered suspicious. Additionally, IP blocking and filtering mechanisms are used to prevent repeated clicks from the same source. These approaches are simple to implement and require less computational effort, making them widely used in earlier advertising platforms.

However, with the rapid advancement in technology, fraudsters have developed sophisticated techniques using automated bots and scripts that can mimic human behaviour. These bots can vary IP addresses, simulate realistic browsing patterns, and bypass traditional detection rules. As a result, existing systems struggle to accurately detect modern and complex fraud patterns.

Limitations of Existing System:-

Despite their simplicity, the existing systems suffer from several significant limitations:

1. Limited Adaptability

Rule-based systems cannot automatically adapt to new and evolving fraud techniques. Any change in fraud patterns requires manual updates to the rules, making the system inefficient in dynamic environments.

2. High False Positives

These systems often incorrectly classify genuine user clicks as fraudulent due to strict or poorly defined rules. This affects user experience and reduces the effectiveness of advertising campaigns.

3. Ineffective Against Advanced Bots

Modern bots are capable of simulating human-like behavior, making it difficult for rule-based systems to distinguish between legitimate users and automated scripts.

4. Lack of Learning Capability

Traditional systems do not learn from historical data or past fraud patterns. They rely only on predefined conditions, which limits their accuracy and performance.

5. Scalability Issues

With the increasing volume of online traffic, rule-based systems struggle to efficiently process and analyze large-scale click data in real time.

6. Poor Detection of Complex Patterns

These systems are unable to detect hidden or non-linear relationships in data, which are often present in sophisticated fraud activities.

2.4 PROPOSED SYSTEM

The proposed system is an intelligent ad click fraud detection framework that utilizes Machine Learning (ML) and Deep Learning (DL) techniques to accurately identify fraudulent advertisement clicks. Unlike traditional rule-based systems, this system is data-driven and capable of learning patterns from historical clickstream data. The system begins with the collection and preprocessing of ad click data, where missing values, duplicate entries, and

noise are removed to ensure data quality. Categorical data is encoded into numerical form, and feature scaling is applied to improve model performance. After preprocessing, important behavioural features such as click frequency, session duration, time intervals between clicks, device information, and geographical patterns are extracted. The processed data is then used to train multiple machine learning models such as Logistic Regression, Support Vector Machine (SVM), Decision Tree, and Random Forest. These models help in identifying common fraud patterns and provide baseline detection performance. In addition to these, deep learning models such as Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) networks are implemented to capture complex, non-linear, and time-dependent patterns in user behaviour. The system integrates predictions from both machine learning and deep learning models to classify ad clicks as fraudulent or legitimate. It supports both batch processing and real-time detection, enabling quick identification of suspicious activities. The system continuously learns from new data, allowing it to adapt to evolving fraud strategies and improve its performance over time.

Furthermore, the system provides a user-friendly interface for uploading datasets, training models, and viewing prediction results. It also includes performance evaluation metrics such as accuracy, precision, recall, and F1-score to assess the effectiveness of the models.

Advantages of the Proposed System:-

The proposed Ad Click Fraud Detection System offers several significant advantages over traditional rule-based systems. These advantages improve the overall efficiency, accuracy, and reliability of fraud detection in online advertising platforms.

1. High Detection Accuracy

The system uses advanced Machine Learning and Deep Learning algorithms, which can accurately identify fraudulent click patterns. This leads to better classification of legitimate and fraudulent clicks.

2. Adaptive and Intelligent Learning

Unlike traditional systems, the proposed system continuously learns from new data and adapts to evolving fraud techniques. This makes it highly effective in dynamic and real-world environments.

3. Reduced False Positives

The system minimizes the chances of incorrectly labeling genuine user clicks as fraudulent. This helps in maintaining a better user experience and improves advertiser trust.

4. Ability to Detect Complex Patterns

Deep learning models such as Neural Networks and LSTM can identify complex, non-linear, and hidden patterns in click data, which are difficult to detect using traditional methods.

5. Real-Time Fraud Detection

The system is capable of analyzing incoming data and providing instant predictions. This helps in quickly identifying and preventing fraudulent activities.

6. Scalability

The system can handle large volumes of ad click data efficiently, making it suitable for high-traffic advertising platforms.

7. Improved System Reliability

By combining multiple models (ML and DL), the system produces more robust and reliable results compared to single-model approaches.

8. Automated Detection Process

The system automates the fraud detection process, reducing the need for manual intervention and minimizing human errors.

9. Better Decision Making

With accurate predictions and performance metrics, advertisers and administrators can make better decisions regarding their ad campaigns.

10. User-Friendly Interface

The system provides an easy-to-use interface for uploading datasets, training models, and viewing results, making it accessible even to non-technical users.

2.5 OBJECTIVES

- To study and analyze different types of ad click fraud such as bot-based, automated, and malicious user-driven attacks.
- To collect and preprocess ad click datasets by removing noise, handling missing values, and encoding categorical features.
- To extract meaningful behavioral features such as click frequency, session duration, and time interval between clicks.

- To implement machine learning algorithms including Logistic Regression, Support Vector Machine, and Random Forest for baseline fraud detection.
- To develop deep learning models such as Deep Neural Networks and Long Short-Term Memory networks to detect complex and time-based fraud patterns.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- System : Pentium IV 2.4 GHz.
- Hard disk : 40GB.
- RAM : 512 Mb.

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows 10
- Language : Python
- Front-End : HTML, CSS
- Database : Mysql , Wamp Server
- Frame Work : Django/Flask

3.SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

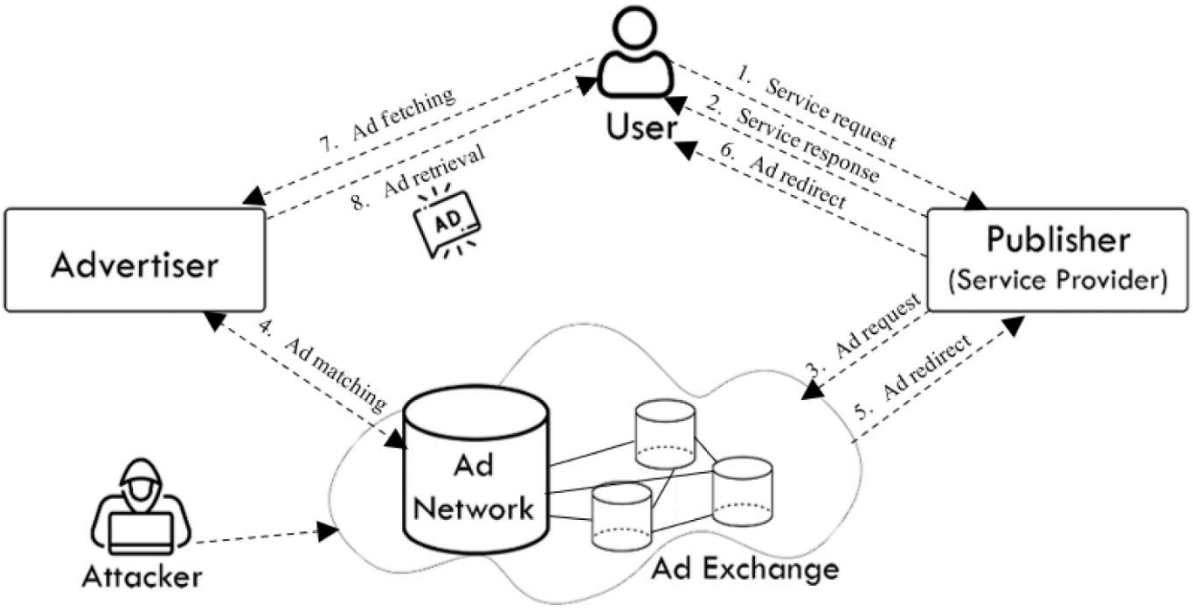

Figure 3.1: System Architecture Of A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads

3.2 DESCRIPTION

The system architecture of the Ad Click Fraud Detection system represents the interaction between different components involved in online advertising, including the User, Publisher, Advertiser, Ad Network, and Ad Exchange. It also considers the presence of attackers who attempt to generate fraudulent clicks.

Components Involved

1. **User**
   The user is the end entity who interacts with online content. The user sends a request to access a website or application and receives advertisements as part of the response.
2. **Publisher(ServiceProvider)**
   The publisher hosts the website or application where advertisements are displayed. It acts as an intermediary between the user and the ad network by requesting ads and displaying them to users.
3. **Advertiser**
   The advertiser creates and provides advertisements. They pay for user clicks under the Pay-Per-Click (PPC) model and rely on accurate click data for campaign performance.
4. **AdNetwork**
   The ad network is responsible for managing advertisements and selecting relevant ads based on user data. It connects advertisers with publishers and performs ad matching.
5. **AdExchange**
   The ad exchange acts as a marketplace where ad requests and responses are processed. It facilitates communication between publishers and ad networks for efficient ad delivery.
6. **Attacker(FraudSource)**
   The attacker represents malicious users or bots that generate fake clicks to manipulate advertising metrics or increase costs.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows within the Ad Click Fraud Detection system. It shows how input data is processed, transformed, and stored while interacting with different components such as users, administrators, and machine learning models. DFD helps in understanding the overall working of the system and how fraudulent clicks are detected.

A Data Flow Diagram comprises Four primary elements:

- External Entities: Represent sources or destinations of data outside the system.
- Processes: Indicate transformations or operations performed on data.
- Data Flows: Depict the movement of data between components.
- Data Stores: Represent where data is stored within the system.

These components are represented using standardized symbols, such as circles for processes, arrows for data flows, rectangles for external entities, and open-ended rectangles for data stores.

Benefits:

The visual representation of the DFD makes it easy to understand how the Ad Click Fraud Detection system operates. It helps both technical and non-technical users to clearly visualize how data moves through the system.

DFDs assist in identifying inefficiencies in data processing, improving system design, and ensuring accurate fraud detection. They also help in maintaining secure data handling by clearly defining how data is accessed and processed within the system.

Applications:

In this project, DFDs are used to model the flow of ad click data from input to prediction. They help in understanding how raw click data is transformed into meaningful insights through machine learning and deep learning models.

DFDs are also useful in software development, system design, and data analysis, as they provide a clear structure of workflows and help identify potential improvements in fraud

detection mechanisms.

Levels of DFD:

DFDs for the Ad Click Fraud Detection system are structured hierarchically:

• **Level 0 (Context Diagram):**
Provides a high-level overview of the system, showing interactions between the user/admin and the fraud detection system.

• **Level 1:**
Breaks down the system into major processes such as dataset upload, data preprocessing, model training, and fraud prediction.

• **Level 2+:**
Provides detailed views of specific processes like feature extraction, model evaluation, and classification, useful for understanding complex operations within the system.

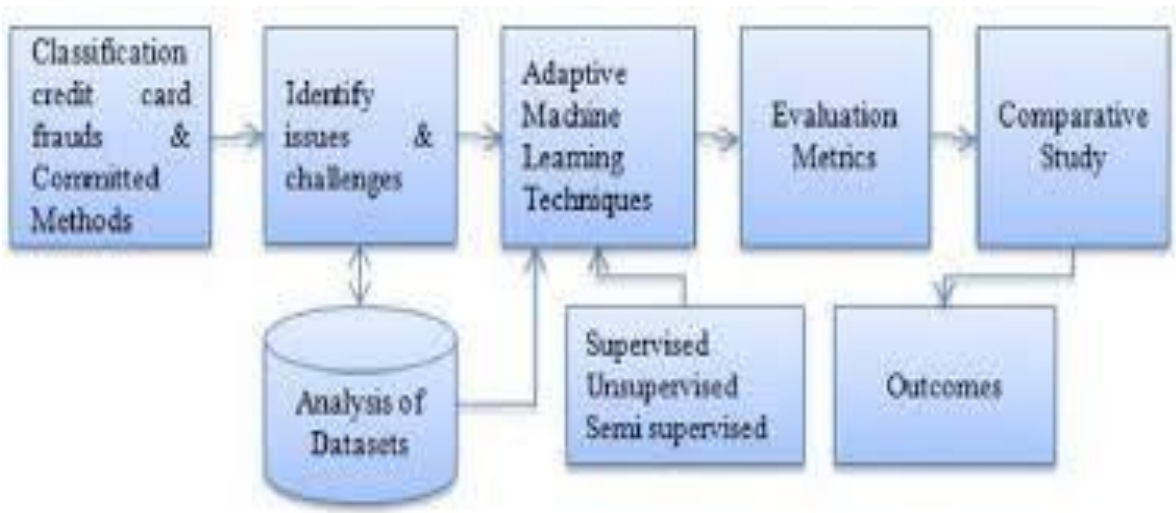

Figure 3.2: Dataflow Diagram of A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of the Ad Click Fraud Detection system involves executing the planned design using Machine Learning and Deep Learning techniques. It includes data collection, preprocessing, feature extraction, model training, and fraud prediction. Proper implementation ensures that the system accurately detects fraudulent clicks and delivers reliable results within a limited time.

The system is developed using efficient algorithms and tools to process large volumes of ad click data. It focuses on achieving high accuracy, reducing false positives, and enabling real-time fraud detection.

4.1 ALGORITHMS USED

Machine Learning Models for Fraud Detection

Machine Learning algorithms are used to analyze structured ad click data and identify patterns that differentiate fraudulent clicks from legitimate ones. Models such as Random Forest and Support Vector Machine (SVM) are implemented to classify clicks based on features like click frequency, time intervals, IP address, and user behaviour.
These models learn from historical data and provide reliable predictions for fraud detection.

Advantages of Machine Learning Models:

• Effective in handling structured data and identifying fraud patterns

• Provide good accuracy with less computational cost

• Easy to implement and interpret

• Suitable for real-time applications

Disadvantages of Machine Learning Models:

• Cannot capture complex and non-linear relationships effectively

• Performance depends on feature selection

• Less effective for highly dynamic and evolving fraud patterns

Deep Learning Models for Advanced Detection :

To overcome the limitations of traditional machine learning, Deep Learning models such as Neural Networks (MLP Classifier) are used. These models can learn complex and hidden patterns in large datasets and improve fraud detection accuracy.
Deep learning models analyze multiple features simultaneously and provide better classification for sophisticated fraud techniques.

Advantages of Deep Learning Models:

• Can detect complex and non-linear fraud patterns

• High accuracy compared to traditional models

• Automatically learns feature representations

• Adapts to evolving fraud techniques

Disadvantages of Deep Learning Models:

• Requires large amounts of data for training

• Computationally expensive

• Longer training time

• Less interpretable compared to ML models

1.Random Forest for Classification

Random Forest is a powerful ensemble machine learning algorithm used for classification and prediction tasks. It works by constructing multiple decision trees during training and combining their outputs to produce a final prediction. In the Ad Click Fraud Detection system, Random Forest plays a crucial role in identifying whether a click is fraudulent or legitimate based on various input features.
The algorithm follows the concept of ensemble learning, where multiple weak learners (decision trees) are combined to form a strong learner. Each decision tree is trained on a random subset of the dataset using a technique called bootstrap sampling, and at each split, a random subset of features is considered. This randomness improves diversity among trees and enhances overall performance.

Working of Random Forest in This Project:

• The input dataset contains features such as click time, IP address, device type, and click frequency

• Multiple decision trees are created using different subsets of data

• Each tree independently predicts whether a click is fraud or legitimate

• Final prediction is made using majority voting among all trees

• The model outputs a stable and accurate classification result

Random Forest is highly effective in detecting ad click fraud because it can capture patterns in structured data and handle variations in user behavior. It provides reliable baseline performance and helps in identifying suspicious clicks with high accuracy.

2. Neural Network (MLP Classifier) for Pattern Learning

The Multi-Layer Perceptron (MLP) is a type of Artificial Neural Network used for learning complex patterns in data. It consists of multiple layers of neurons, including an input layer, one or more hidden layers, and an output layer. In the Ad Click Fraud Detection system, the MLP classifier is used to identify complex and non-linear relationships in click data that traditional models may fail to detect.

MLP works using a process called **forward propagation**, where input data passes through layers of neurons, and each neuron applies weights and activation functions to generate output. The model is trained using **backpropagation**, where errors are minimized by adjusting weights iteratively.

Working of MLP in This Project:

• Input layer receives features such as click time, user behavior, and device information
• Data passes through hidden layers where complex patterns are learned
• Activation functions (ReLU, sigmoid) are applied to introduce non-linearity
• Output layer predicts whether the click is fraud or legitimate
• Model improves over time by minimizing prediction error

Advantages of MLP Classifier:

• Processes multiple input features simultaneously
• Learns complex and non-linear relationships in click data
• Provides higher accuracy compared to traditional ML models
• Automatically performs feature learning
• Suitable for large-scale and complex datasets

Disadvantages of MLP Classifier:

• Requires large datasets for effective training
• Computationally intensive and slower compared to simple models
• Difficult to interpret (black-box model)
• Needs proper tuning of hyperparameters

Role in Fraud Detection:

The MLP classifier enhances fraud detection by identifying hidden and complex patterns in user behaviour that are not easily captured by traditional algorithms. It improves overall system accuracy and helps detect sophisticated fraud techniques used by bots and attackers.

In the proposed system, Random Forest and MLP classifiers work together to provide an efficient and accurate fraud detection mechanism. While Random Forest offers stability and reliability, MLP improves accuracy by capturing complex patterns, making the system robust and effective.

Nowadays, online advertising platforms are widely used by businesses to promote products and services. However, malicious users and automated bots generate fake clicks to manipulate ad performance and increase advertising costs.
To address this issue, this project implements a fraud detection system using Machine Learning and Deep Learning techniques. The system processes ad click data, extracts important features, and uses models like Random Forest and Neural Networks to classify clicks as fraudulent or legitimate.
The proposed system improves accuracy compared to traditional methods and provides a reliable solution for detecting and preventing ad click fraud in online advertising platforms.

4.2 SAMPLE CODE

```
from django.shortcuts import render, redirect

from django.http import HttpResponse

from django.core.files.storage import FileSystemStorage

import pandas as pd

import numpy as np

import os

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.ensemble import RandomForestClassifier

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score

# ----------------------------

# GLOBAL MODELS (In-Memory)

# ----------------------------

rf_model = None

dl_model = None

scaler = None

label_encoders = {}
```

```
# ----------------------------
# HOME PAGE
# ----------------------------
def home(request):
    return render(request, "home.html")
# ----------------------------
# DATASET UPLOAD
# ----------------------------
def upload_dataset(request):
    if request.method == "POST":
        file = request.FILES["dataset"]
        fs = FileSystemStorage()
        filename = fs.save(file.name, file)
        request.session["dataset"] = filename
        return redirect("train_models")
    return render(request, "upload.html")



# ----------------------------
```

```
# DATA PREPROCESSING

# ----------------------------

def preprocess_data(df):

    global label_encoders, scaler

    df = df.dropna()

    for col in df.columns:

        if df[col].dtype == "object":

            le = LabelEncoder()

            df[col] = le.fit_transform(df[col])

            label_encoders[col] = le

    X = df.drop("Fraud", axis=1)

    y = df["Fraud"]

    scaler = StandardScaler()

    X = scaler.fit_transform(X)

    return train_test_split(X, y, test_size=0.2, random_state=42)

# ----------------------------

# TRAIN ML & DL MODELS

# ----------------------------

def train_models(request):

    global rf_model, dl_model
```

```
file = request.session.get("dataset")

if not file:

    return HttpResponse("No dataset uploaded")

df = pd.read_csv("media/" + file)

X_train, X_test, y_train, y_test = preprocess_data(df)

# -------- ML MODEL --------

rf_model = RandomForestClassifier(n_estimators=100)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

rf_acc = accuracy_score(y_test, rf_pred)

# -------- DL MODEL (MLP) --------

dl_model = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500)

dl_model.fit(X_train, y_train)

dl_pred = dl_model.predict(X_test)

dl_acc = accuracy_score(y_test, dl_pred)

return render(

    request,

    "train.html",

    {

        "rf_acc": round(rf_acc * 100, 2),
```

```
            "dl_acc": round(dl_acc * 100, 2),
        },
    )
# ----------------------------
# FRAUD PREDICTION PAGE
# ----------------------------
def predict_fraud(request):
    result = None
    if request.method == "POST":
        input_data = request.POST.dict()
        input_data.pop("csrfmiddlewaretoken")
        df = pd.DataFrame([input_data])
for col in df.columns:
            if col in label_encoders:
                df[col] = label_encoders[col].transform(df[col])


        df = scaler.transform(df)
        rf_result = rf_model.predict(df)[0]
        dl_result = dl_model.predict(df)[0]
        if rf_result == 1 or dl_result == 1:
```

```
        result = "🚨 Fraudulent Click Detected"

    else:

        result = "✅ Legitimate Click"

    return render(request, "predict.html", {"result": result})

# ----------------------------

# REPORT PAGE

# ----------------------------

def report(request):



 return render(request, "report.html")
```

Pseudo Code:

```
BEGIN

  // Step 1: System Initialization

  Initialize Web Application

  Initialize Database

  Initialize Machine Learning and Deep Learning Models



  // Step 2: User Authentication
```

```
  IF user requests login THEN
    Validate username and password
    IF credentials valid THEN
      Grant access
    ELSE
      Deny access
    ENDIF
  ENDIF
// Step 3: Dataset Upload
  IF admin uploads dataset THEN
    Store dataset in database
  ENDIF
  // Step 4: Data Preprocessing
  Load ad click dataset
  Remove missing values
  Remove duplicate records
  Encode categorical attributes
  Normalize numerical features


  // Step 5: Feature Extraction
```

```
  Extract behavioral features:
      - Click frequency
      - Session duration
      - Time gap between clicks
      - Device and browser patterns
// Step 6: Model Training
  Split dataset into training and testing sets
  Train Machine Learning Model (Random Forest / SVM)
  Train Deep Learning Model (DNN / LSTM)


  // Step 7: Fraud Detection
  FOR each ad click record DO
      Predict using ML model
      Predict using DL model


      IF ML prediction OR DL prediction indicates fraud THEN
          Label click as FRAUDULENT
      ELSE
          Label click as LEGITIMATE
      ENDIF
```

```
  END FOR

  // Step 8: Performance Evaluation
  Calculate Accuracy
  Calculate Precision, Recall, F1-Score

// Step 9: Alert Generation
  IF fraud rate exceeds threshold THEN
     Generate alert for administrator
  ENDIF

   // Step 10: Reporting
  Generate fraud analysis report
  Display results on dashboard

// Step 11: Model Update
  Periodically retrain models with new data
END
```

5.RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

GUI / Login Interface:

The below screen shows the main interface of the Ad Click Fraud Detection system with login functionality. Users can select Admin/User, enter credentials, and click login to access the system.

Figure 5.1 : GUI/Login Interface of the system

Admin Dashboard / Remote Users View :

The below screen shows the admin dashboard where various operations like dataset upload, training, and prediction can be performed. It also displays the list of all registered remote users with their details.

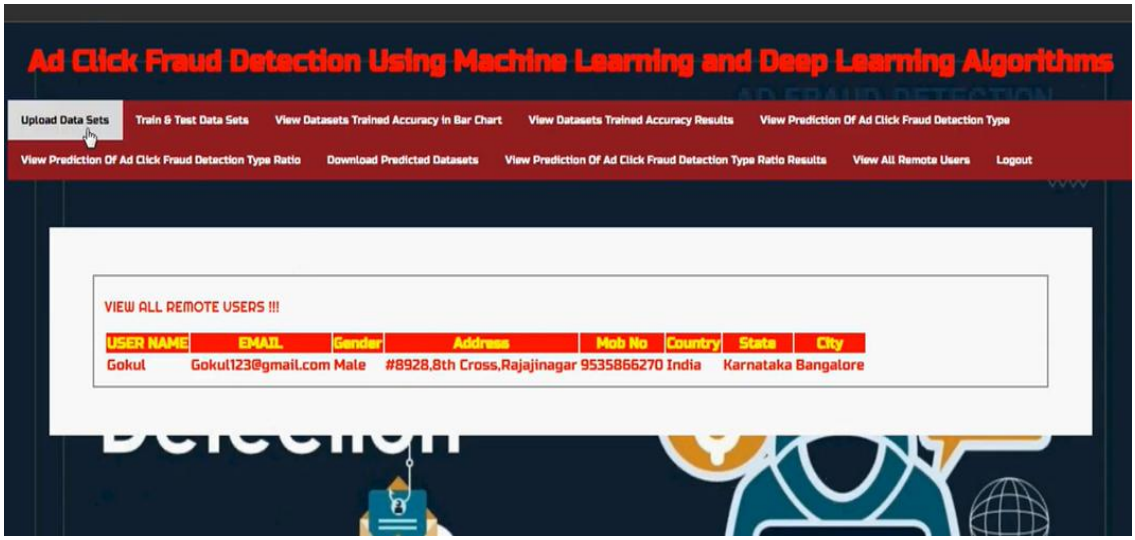

Figure 5.2: Admin dashboard showing remote users information

Dataset Upload Interface:

The below screen shows the dataset upload process where the admin selects files using the browse option. After selecting the dataset, it is uploaded into the system for further processing and training.

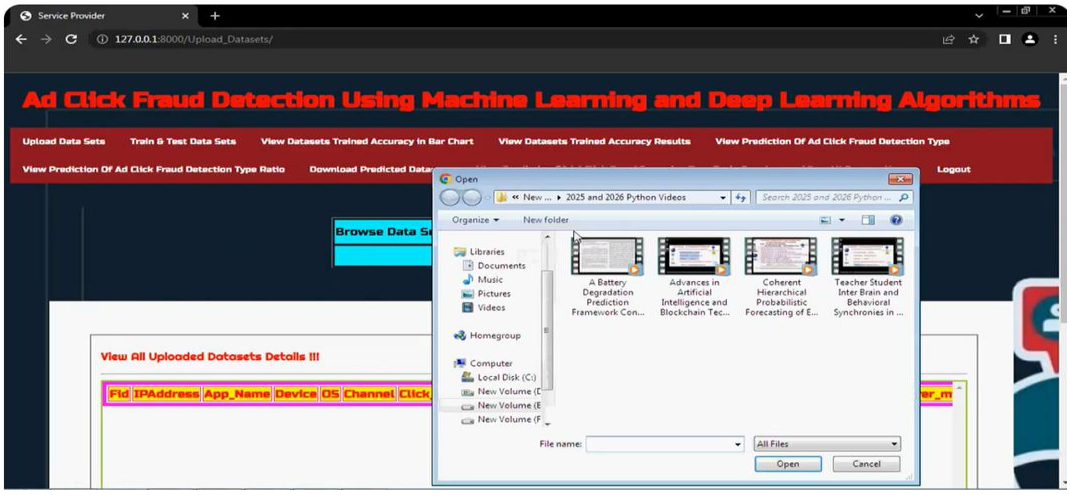

Figure 5.3 : Dataset upload interface of the system

Dataset Details View Interface:

The below screen displays the uploaded dataset details in a tabular format. It shows various attributes such as timestamps, numerical values, and feature columns used for training the model. This interface helps the admin verify and analyze the dataset before processing.

Figure 5.4 : Dataset details view interface of the system

Model Training Results Interface:

The below screen displays the accuracy results of different Machine Learning and Deep Learning models used in the system. It shows a comparison of models such as Random Forest, KNN, Gradient Boosting, Logistic Regression, and Deep Neural Network based on their performance. This helps in identifying the best model for fraud detection.

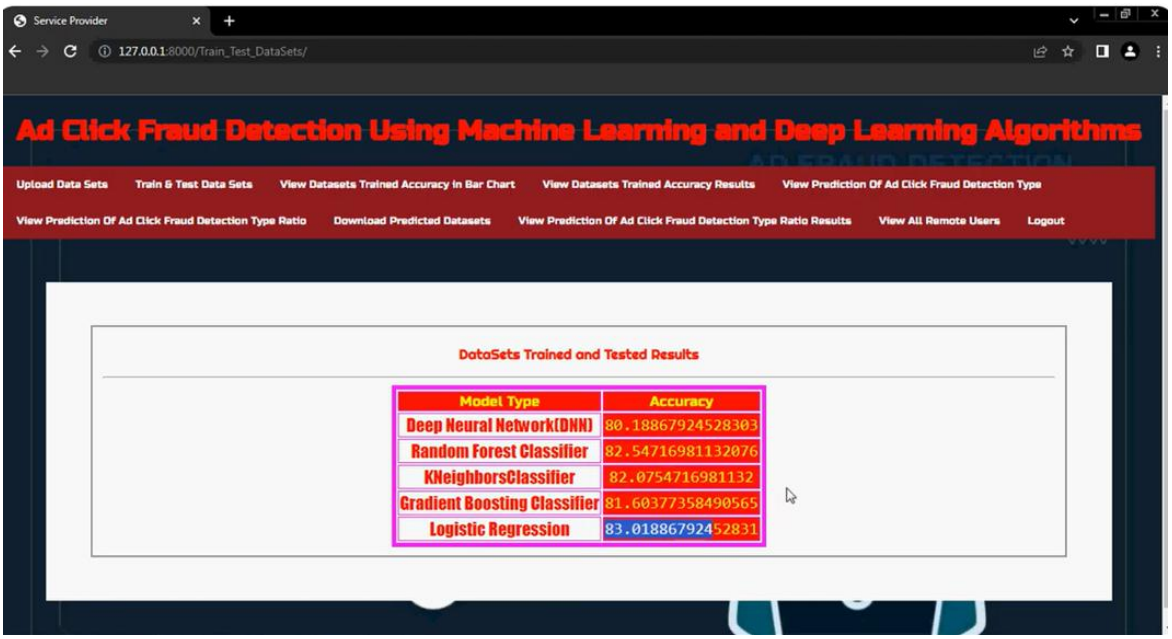

Figure 5.5 : Model training results interface of the system

Model Accuracy Bar Chart Interface:

The below screen illustrates a bar chart representing the accuracy of different models used in the system. It visually compares the performance of algorithms such as Random Forest, KNN, Gradient Boosting, Logistic Regression, and Deep Neural Network. This graphical representation helps in easily identifying the most accurate model.

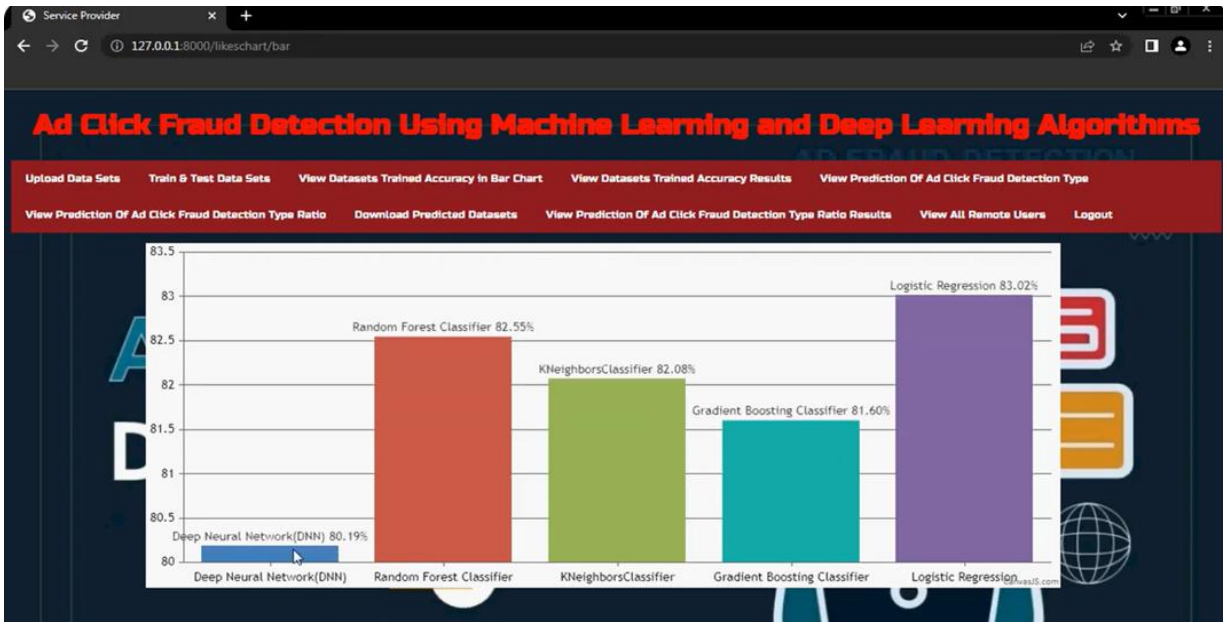

Figure 5.6 : Model accuracy bar chart interface of the system

Model Accuracy Line Chart Interface:

The below screen displays a line chart representing the accuracy trends of different Machine Learning and Deep Learning models. It shows how each model performs in comparison, including Random Forest, KNN, Gradient Boosting, Logistic Regression, and Deep Neural Network. This visualization helps in analyzing performance variations across models.

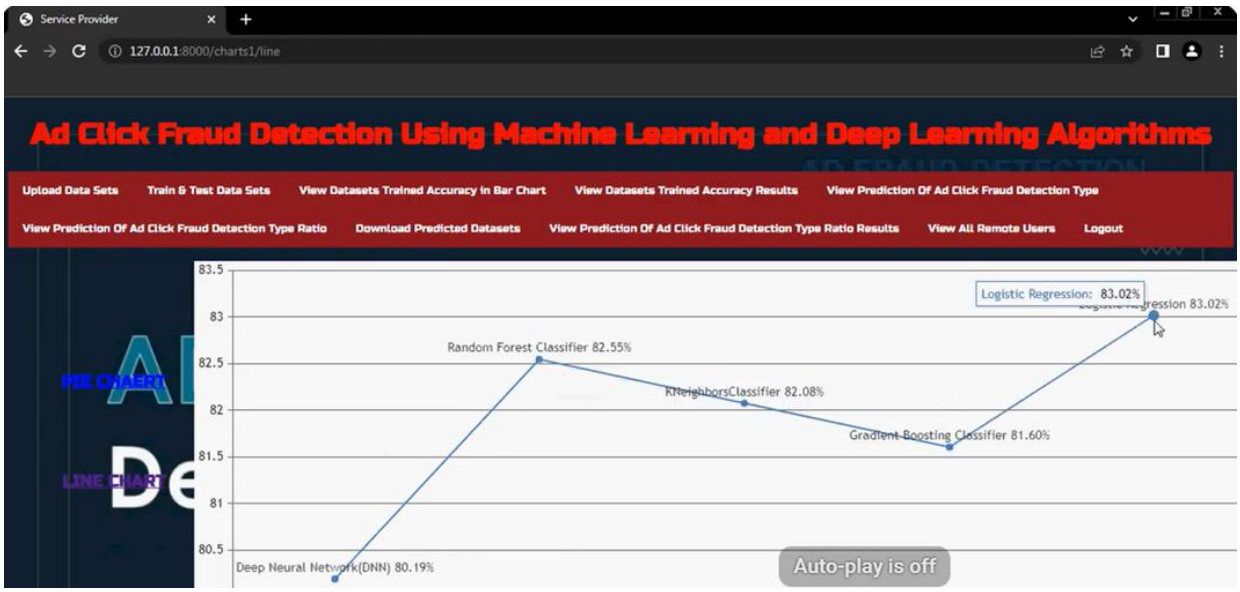

Figure 5.7 : Model accuracy line chart interface of the system

Ad Click Fraud Detection Type Ratio Interface:

The below screen displays the ratio of predicted ad click fraud results. It shows the percentage of "Fraud Detected" and "Fraud Not Detected" instances after model prediction. This helps in understanding the overall distribution of fraudulent and genuine clicks.

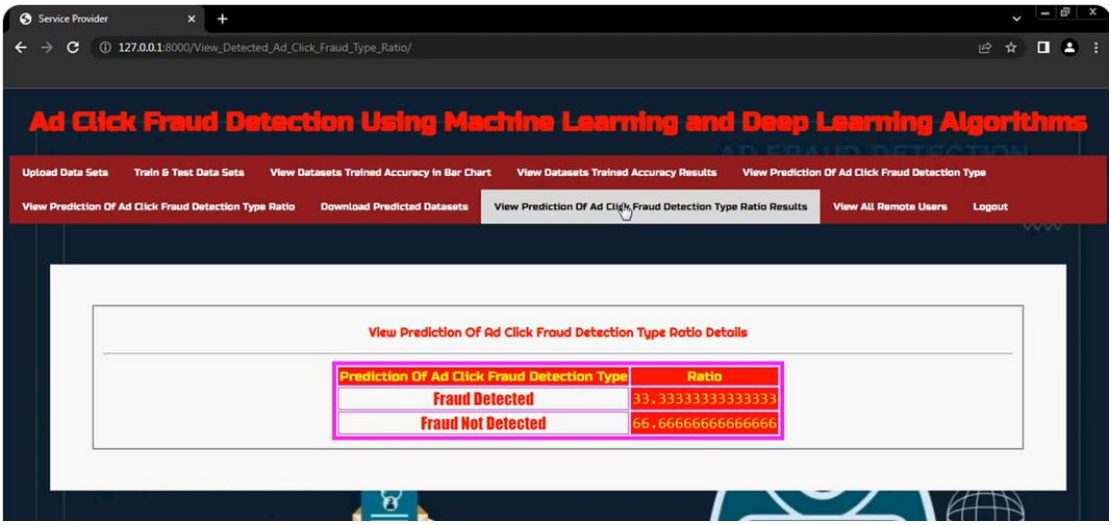

Figure 5.8 : Prediction ratio interface of ad click fraud detection system

User Registration Interface:

The below screen shows the registration page where new users can create an account. It collects details such as username, email, password, gender, address, and contact information. This interface enables users to access the system securely.

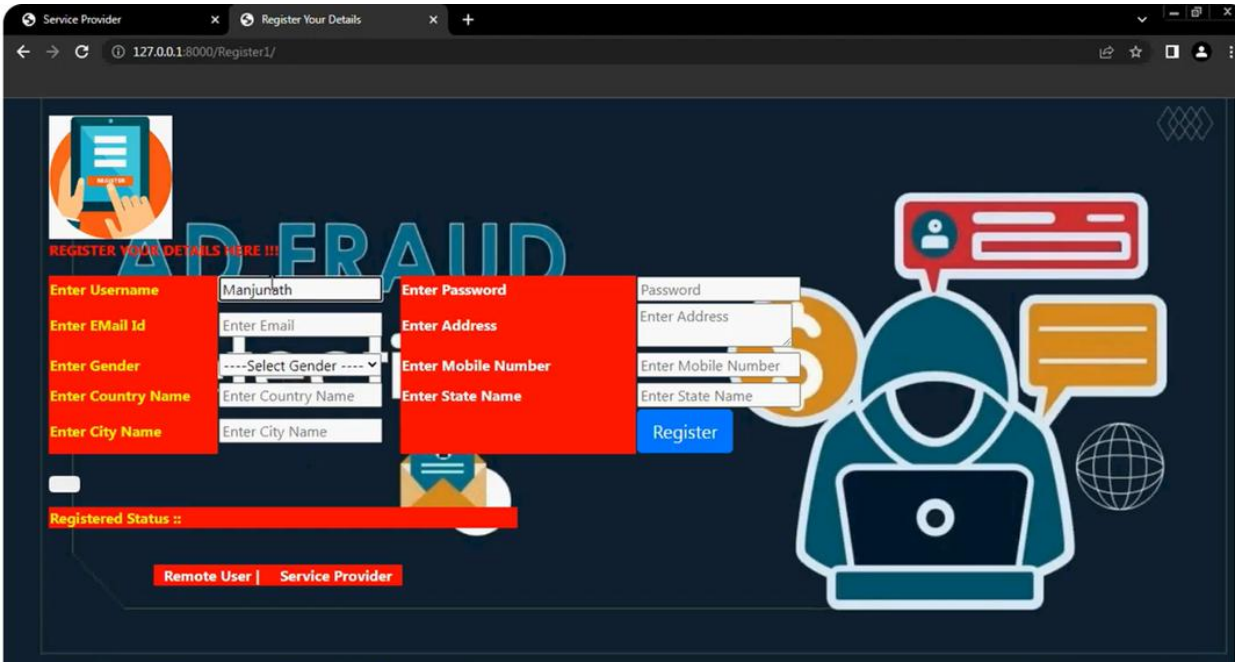

Figure 5.9 : User registration interface of the system

User Login Interface:

The below screen displays the login page for existing users. Users enter their username and password to access the system functionalities. It also provides options for service provider login and new user registration.

The interface ensures secure authentication before granting access to the system. It helps in managing authorized users and prevents unauthorized access. After successful login, users are redirected to the main dashboard.

Figure 5.10 : User login interface of the system

Ad Click Fraud Detection Input Interface:

The below screen shows the input interface where users enter required dataset features such as FID, App Name, OS, IP Address, Device, and other behavioral attributes. These inputs are used by the system to analyze ad click activity. The form ensures all necessary parameters are provided for accurate prediction.

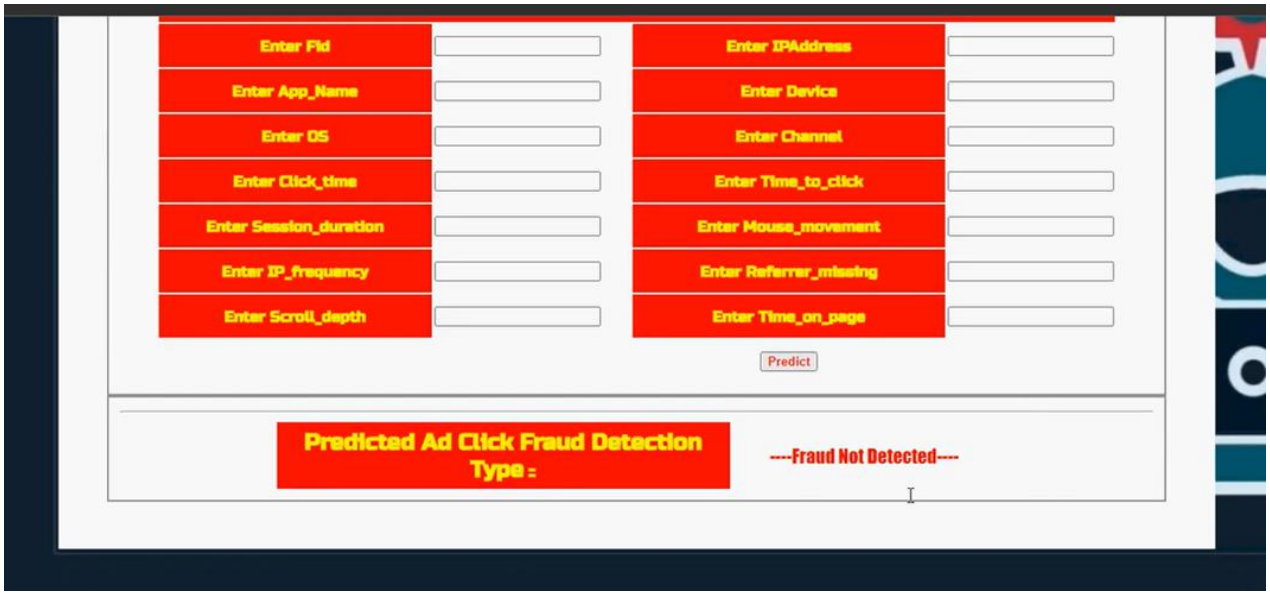

Figure 5.11 : Input interface for ad click fraud detection system

Ad Click Fraud Prediction Result Interface:

The below screen shows the prediction result interface after submitting the input data. Once the user clicks the "Predict" button, the system processes the entered values using the trained model and displays the result at the bottom. The output section clearly indicates whether the ad click is fraudulent or not (e.g., "Fraud Not Detected"). This interface helps users quickly interpret the model's decision based on the provided inputs.

Figure 5.12 : Prediction result interface of ad click fraud detection system

6.VALIDATION

6. VALIDATION

The validation of this project is performed through systematic testing and evaluation to ensure the accuracy and reliability of the ad click fraud detection system. The process includes dataset validation, preprocessing verification, model training evaluation, and real-time prediction testing. A structured validation approach helps in minimizing false predictions and improving the system's capability to distinguish between fraudulent and legitimate ad clicks.

6.1 INTRODUCTION

Initially, the uploaded dataset is divided into training and testing sets using an 80-20 split. The training data is used to train both the Machine Learning model (Random Forest) and the Deep Learning model (MLP Classifier), while the testing data is used to evaluate their performance on unseen data. This ensures that the models can generalize effectively and avoid overfitting.

The system applies preprocessing techniques such as handling missing values, encoding categorical features using Label Encoding, and normalizing numerical data using Standard Scaler. These steps ensure that the data is in a suitable format for accurate model training and prediction.

The performance of the system is evaluated using key metrics such as accuracy score, which is calculated for both Random Forest and Deep Learning models. The results are displayed to compare model performance and ensure that the system provides reliable predictions. Higher accuracy indicates better fraud detection capability.

Finally, real-time validation is performed through the prediction interface, where user-input data is processed by both models. If either model detects fraud, the system labels the click as fraudulent; otherwise, it is marked as legitimate. This combined approach improves detection accuracy and ensures robust performance in real-world scenarios.

6.2 TEST CASES

TABLE 6.2.1: UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case Description	Output
1.	Dataset Upload	To upload dataset for model training	User uploads the dataset file for fraud detection	Dataset successfully loaded.

TABLE 6.2.2: CLASSIFICATION (FRAUD DETECTION)

Test case ID	Test case name	Purpose	Input	Output
1	Fraud Detection Test 1	To check if system detects legitimate clicks	Normal user click data	Legitimate Click
2	Fraud Detection Test 2	To check if system detects fraudulent clicks	Suspicious click data.	Fraudulent Click Detected

7.CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objective of detecting fraudulent ad clicks using Machine Learning and Deep Learning techniques. The system was designed and implemented effectively, enabling accurate classification of ad clicks as fraudulent or legitimate. By integrating multiple models and preprocessing techniques, the project provides reliable predictions and valuable insights into user behavior.

Looking ahead, the system has strong potential for further enhancement by incorporating advanced algorithms, real-time data processing, and improved scalability. Future developments will focus on increasing detection accuracy, reducing false predictions, and adapting to evolving fraud patterns. These improvements will ensure the system remains efficient, robust, and applicable in real-world digital advertising environments.

7.1 PROJECT CONCLUSION

This project presents an effective ad click fraud detection system using a hybrid approach that combines Machine Learning (Random Forest) and Deep Learning (MLP Classifier). The system processes user input data through preprocessing steps such as data cleaning, label encoding, and feature scaling to ensure high-quality input for model training. Both models are trained and evaluated using accuracy metrics, providing a comparative analysis of their performance.

The proposed system improves fraud detection by using dual-model prediction, where a click is marked as fraudulent if either model detects suspicious activity. This approach enhances detection reliability and reduces the chances of missing fraudulent clicks. The system also provides a user-friendly interface for dataset upload, prediction, and result visualization.

Overall, the project demonstrates the practical application of AI in digital advertising security. It helps in identifying abnormal click patterns, preventing revenue loss, and improving trust in online advertising systems. The modular design allows easy scalability and integration with real-time applications, making it suitable for future deployment.

7.2 FUTURE ASPECTS

The proposed ad click fraud detection system shows promising results; however, there is significant scope for further improvements. Future enhancements can focus on improving model performance, scalability, and real-time capabilities.

Some key future aspects include:

- **Real-Time Fraud Detection:** Implementing real-time streaming data analysis to detect fraud instantly.
- **Advanced Deep Learning Models:** Using models like LSTM or CNN for better pattern recognition in user behavior.
- **Big Data Integration:** Handling large-scale datasets using technologies like Hadoop or Spark.
- **Explainable AI (XAI):** Providing clear explanations for predictions to improve transparency and trust.
- **Adaptive Learning:** Continuously updating models with new data to detect evolving fraud patterns.
- **Integration with Ad Platforms:** Deploying the system in real-world advertising networks for live monitoring.
- **Enhanced Feature Engineering:** Incorporating more behavioral features such as geolocation patterns and click timing.

These future improvements will make the system more accurate, scalable, and efficient, ensuring better protection against ad click fraud in dynamic digital environments.

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8.BIBLIOGRAPHY

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8.2 GITHUB LINK

https://github.com/nithinvalaboju123/MAJOR-PROJECT-batch 21

8.3 PUBLICATIONS

The research work carried out in this project has been successfully accepted for presentation and publication at the International Conference on Multidisciplinary Technology, Innovation, and Management Science (ICMTIMS '26). The paper, titled *"A Data-Intelligent Framework for Detecting Manipulated Click in Online Ads"*, focuses on the proposed system for detecting fraudulent ad clicks using Machine Learning and Deep Learning techniques. This work highlights the application of artificial intelligence in identifying suspicious click patterns and preventing digital advertising fraud, thereby contributing to secure and reliable online advertising systems.

The acceptance of this paper reflects the novelty, technical strength, and practical relevance of the proposed approach in the fields of machine learning, data analytics, and cybersecurity. The study demonstrates how a hybrid framework combining models such as Random Forest and Deep Learning can effectively detect fraudulent click behaviour, improving the accuracy and reliability of fraud detection systems.

Furthermore, the conference provides a valuable platform to present the research work, receive constructive feedback from reviewers, and interact with experts and researchers from various domains. The peer-review process also highlighted areas for improvement, including better explanation of dataset details, feature selection, and comparative analysis with existing methods. This publication enhances the academic value and global visibility of the research, emphasizing its contribution toward building secure and trustworthy digital advertising environments.

8.3.1 CONFERENCE CERTIFICATIONS

The following conference certifications were obtained as part of participation and presentation in an international conference. These certifications validate active involvement in academic research, dissemination, and professional development.

Details:

- **Conference:** International Conference on Multidisciplinary Technology, Innovation, and Management Science (ICMTIMS 2026)
- **Date:** March 28, 2026
- **Type:** Paper Presentation
- **Paper ID:** ICMTIMS – 251
- **Title:** A Data-Intelligent Framework for Detecting Manipulated Click in Online Ads

Acceptance Notification – ICMTIMS '26 Inbox ×

ICMTIMS conference <icmtimsconference@gmail.com> Thu 26 Mar, 15:18 (8 days ago)
to me, nithinvalaboju, Sriram, bhavya.cse, sn.suma05

Dear Author **P Bhavani Priya, Valaboju Nithin, G. Sriram, D Sandhya Rani, Suma S, B Bhavya**

We are pleased to inform you that your paper titled **"A Data-Intelligent Framework for Detecting Manipulated Click in Online Ads"** with Paper ID: **ICMTIMS - 251** has been officially accepted for presentation at the **International Conference on Multidisciplinary Technology, Innovation, and Management Science (ICMTIMS '26)** after a successful peer-review process.

Date of Acceptance: 26.03.2026

Your paper has been evaluated by the Technical Review Committee and found suitable for publication and presentation. Kindly ensure that all reviewer-suggested corrections (if any) are incorporated into the camera-ready version before final submission. The official camera-ready template will be shared with you shortly.

Review Comments:

Comment 1:

The paper proposes a robust data-intelligent framework combining machine learning and deep learning for ad click fraud detection, but the manuscript can be improved by providing clearer explanation of dataset details, feature selection process, and reproducibility of experiments

Comment 2:

The experimental results demonstrate strong performance of deep learning models; however, the paper would benefit from deeper comparative analysis with existing methods and more discussion on limitations such as computational cost and data imbalance

Important Notes:

- The final camera-ready paper must be submitted as per the official conference template before the specified deadline.
- ICMTIMS '26 proceedings are indexed in **Scopus-indexed outlets**.
- At least one author must register and present the paper at the conference for inclusion in the final proceedings.

Reply Forward

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CERTIFICATE

of Presentation on 27 & 28 March 2026

This Certificate is Proudly Presented To

Suma S

A Data-Intelligent Framework For Detecting Manipulated Click In Online Ads

This is to certify that the paper was presented at the International Conference on Multidisciplinary Technology, Innovation and Management Science (ICMTIMS-2026), organized by St. Claret College Autonomous, Bengaluru.

Rev. Thomas V. Thennadiyil, CMF Principal St. Claret College Autonomous, Bengaluru

Prof. Cassandra Ranee HoD - Dept. Of Management (UG) St. Claret College Autonomous, Bengaluru

Dr. Arijit Roy HoD - PG Dept. Of Management St. Claret college Autonomous,Bengaluru

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Dr. Arijit Roy HoD - PG Dept. Of Management St. Claret college Autonomous,Bengaluru

Ms. Subiksha.S Industry Partner, NTL Technology, TamilNadu.